UTAH ACCESS

Analysis of Cycling Comfort, Efficiency, and Sustainable Solutions using GIS

Presenters: Alex Fielder, Elijah Manwill
Mentor: Dr. Aishwarya Chandrasekaran

S.J. & Jessie E. Quinney
College of Natural Resources
UtahStateUniversity
Department of Environment and Society

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INTRODUCTION

Barrier to adopting cycling
- Comfort, **perceived safety**
- Lack of Cycling Infrastructure – **can be fatal**
- Traffic

Other methods
- Lack critical factors (Slope, Trails)
- Complicated to Source Data
- Need for Simplicity

1-4 Metric
- Borrowed from Level of Traffic Stress (LTS) [https://doi.org/10.3141/2587-06]

WASATCH FRONT NEWS

There were nearly 550 bike-related crashes on Utah roads last year. And the helmet use rate was 'abysmal'

by: Jonathon Sharp
Posted: Jul 18, 2024 / 04:32 PM MDT
Updated: Jul 18, 2024 / 04:33 PM MDT

NEWS > LOCAL NEWS

2022 was deadliest year for bike fatalities, injuries in Utah since data tracking began

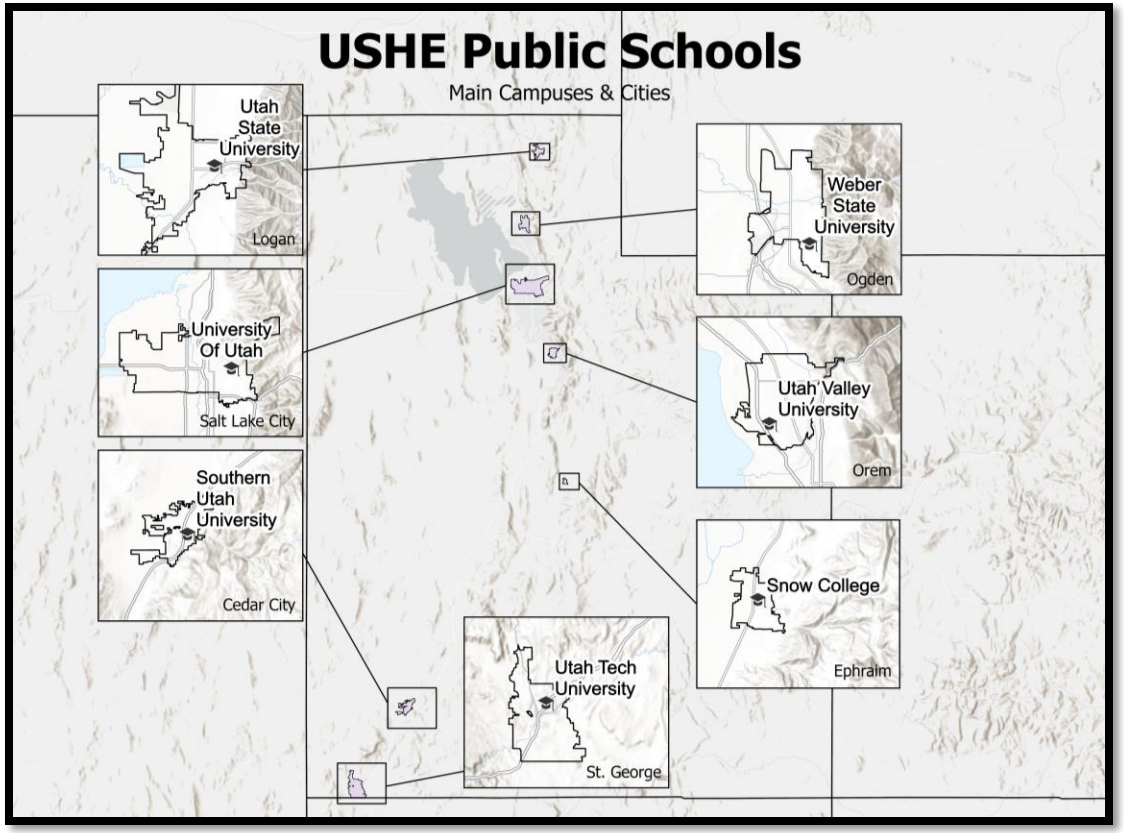
Photo by: Stewart family

Adam Stewart at the Utah High School Cycling State Championship before his fall.

Posted 2:10 PM, May 19, 2023

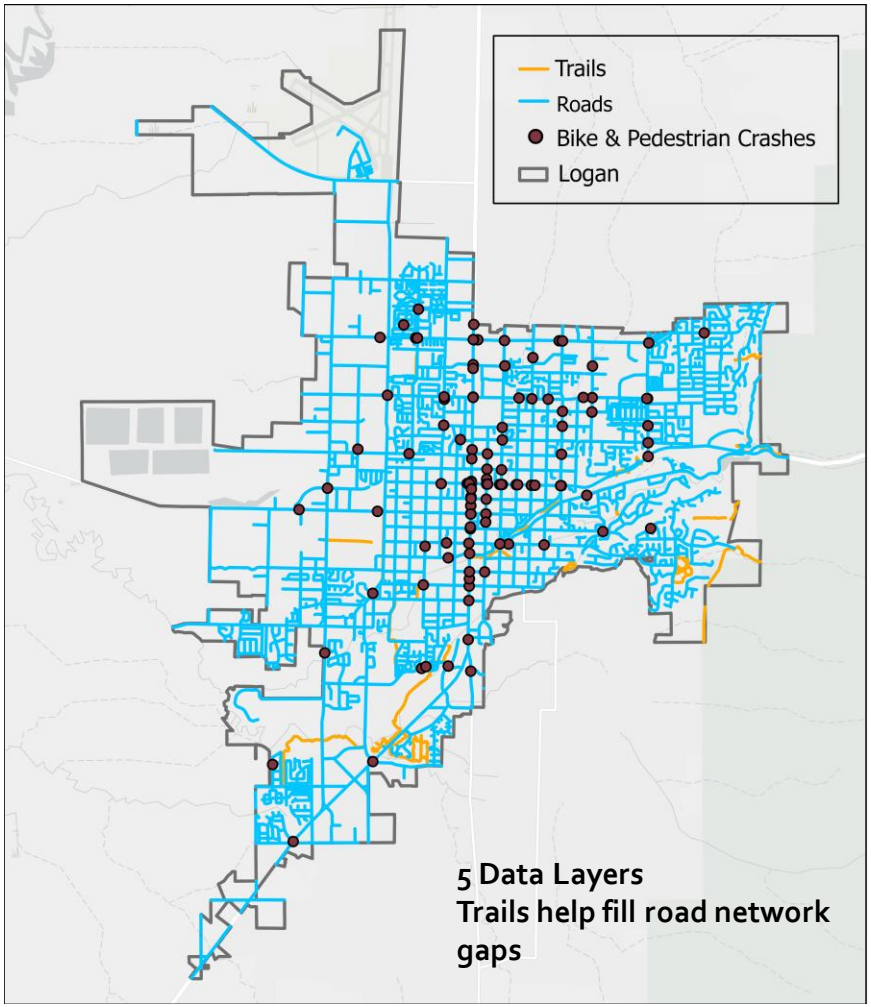
STUDY AREA

- College students
  - Transit costs
  - Campus design
  - Willingness
- Utah System of Higher Edu (**USHE**) Public Schools
  - Primary Campuses
  - Large range of city size and campus population
  - Student focused

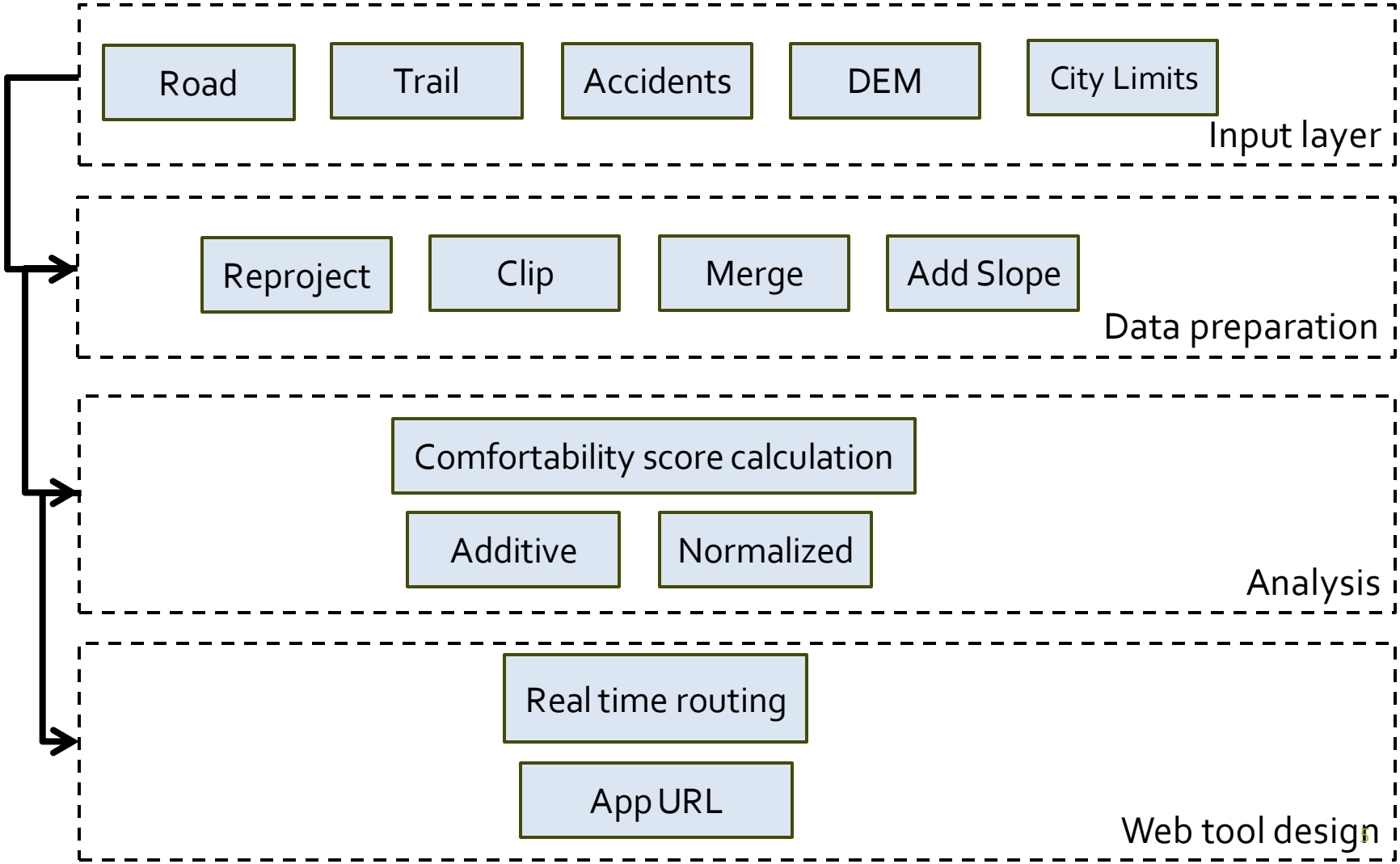


DATA DESCRIPTION

Data Name	Relevant Fields	Source	Type
Roads	Bike Lane Type, Speed Limit, Annual average daily traffic	UGRC	Vector - Line
Trails	Trail Type	UGRC	Vector - Line
City Boundaries	City	UGRC	Vector - Polygon
Crash Data	Cyclists and Pedestrians	Utah Gov Open Data	CSV
10m DEM		UGRC	Raster



METHODS


Road
Trail
Accidents
DEM
City Limits
Input layer
Reproject
Clip
Merge
Add Slope
Data preparation
Comfortability score calculation
Additive
Normalized
Analysis
Real time routing
App URL
Web tool design

FINAL DATA LAYERS

Bike Lane Types

Speed Limit

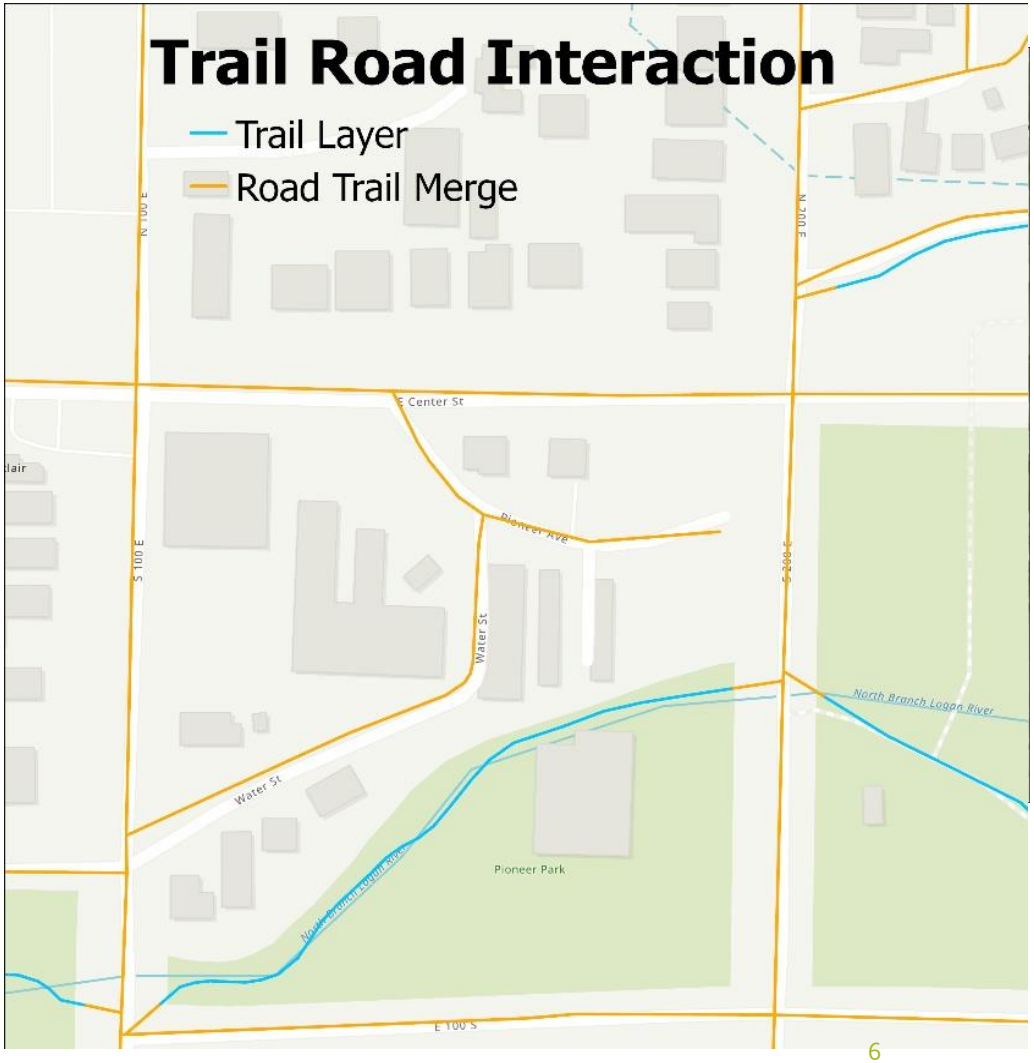
AADT

(Average Annual Daily Traffic)

Slope (Energy)

Crime (Crashes)

Trails



BIKING COMFORTABILITY SCORING

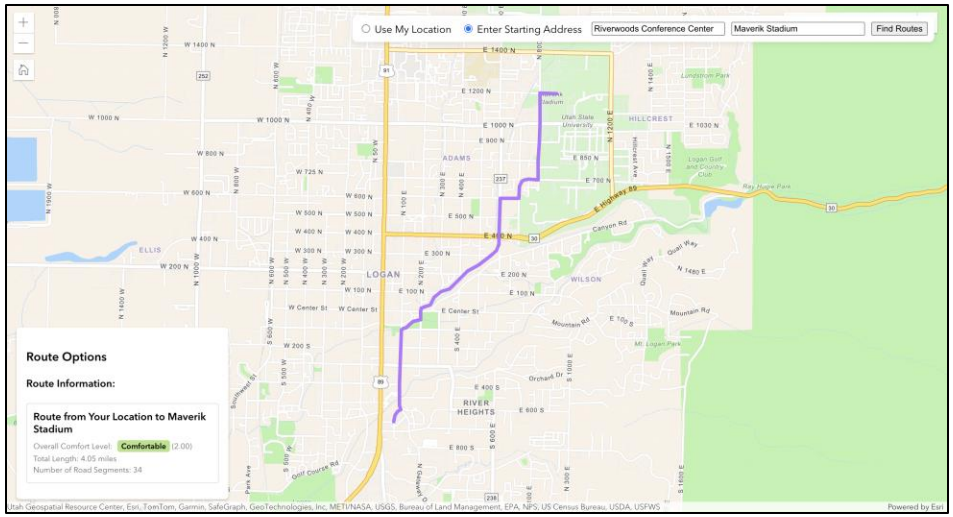
Manual scoring Method	Normalized Method

• Each field was scored from 1 to 4 based on **perceived comfortability** and LTS criteria
• After manual scoring, **all the attributes are added.**
• Final score range – 1- 4
  • 1 - highest comfort
  • 4 - lowest comfort
	• Normalizes each fields value 0-1 (except bike lanes) and each field is added together
	• Each field equally contributes to the final comfortability score
	  • **Bike lanes improve scores but do not penalize**
	• **Final comfortability score** calculated using **quantile break condition**
	• Removes observer bias in scoring

REALTIME ROUTING AND WEBTOOL

Key Features

- Interactive Map Interface
- Route Planning Based on Comfort Scores as Opposed to Shortest Time
- Real Time Route Calculation
- User Friendly, Intuitive Design
- Tool Rated with Comfortability Scores from 1-4, with 1 Being the Most Comfortable



TECHNICAL ARCHITECTURE

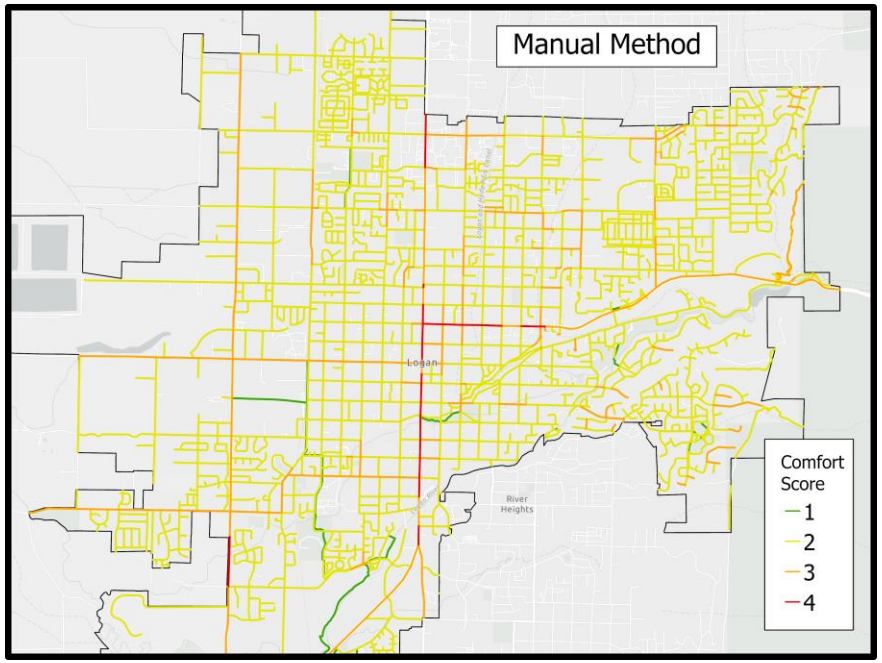
Front End

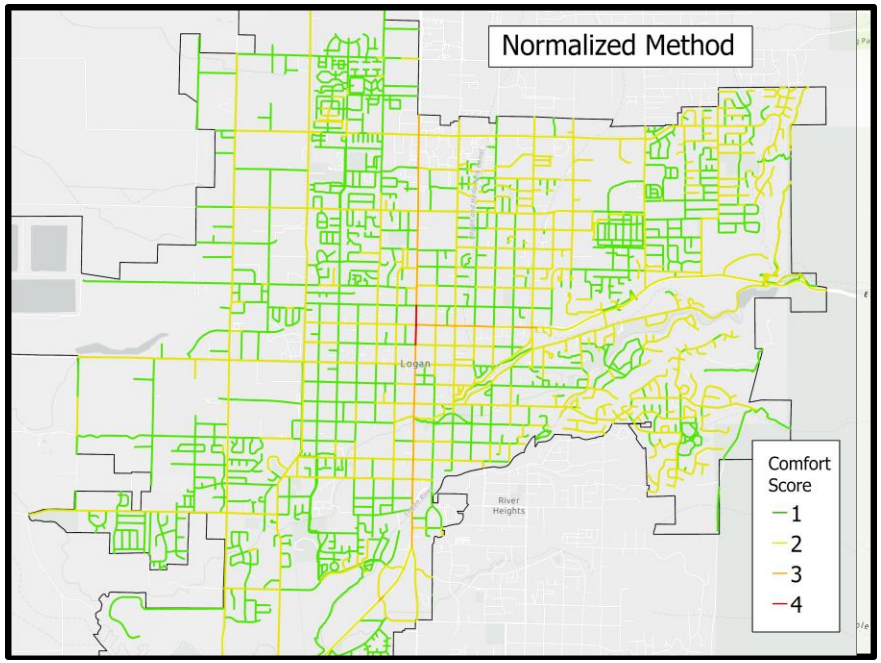
- ArcGIS JavaScript API for Mapping
- Modern HTML5/CSS3 Interface

Back End

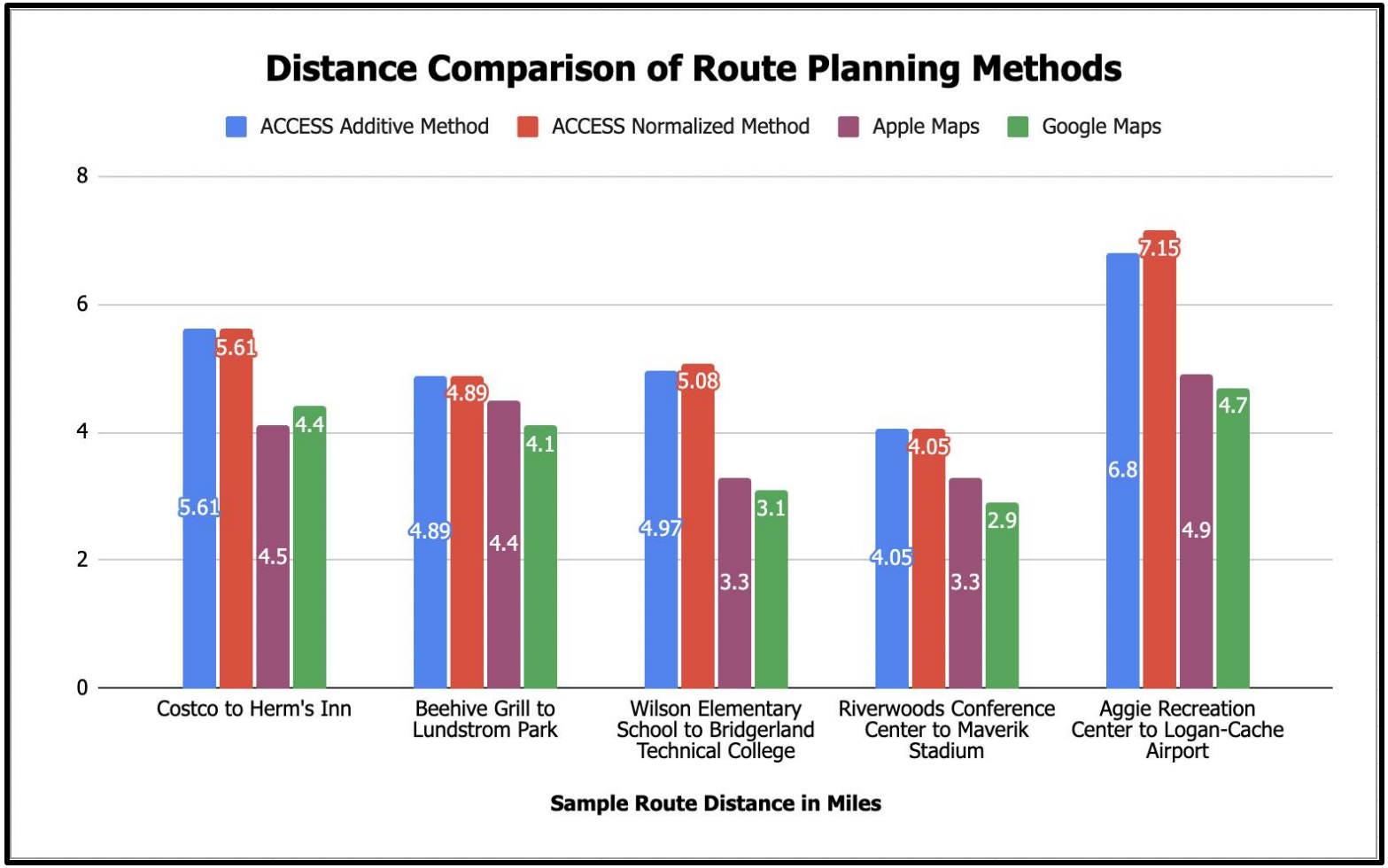
- Flask (Python) Server
- GeoPandas for Processing, OpenStreetMap for geocoding
- Custom Comfort Scoring Algorithm

RESULTS: VISUAL COMPARISON





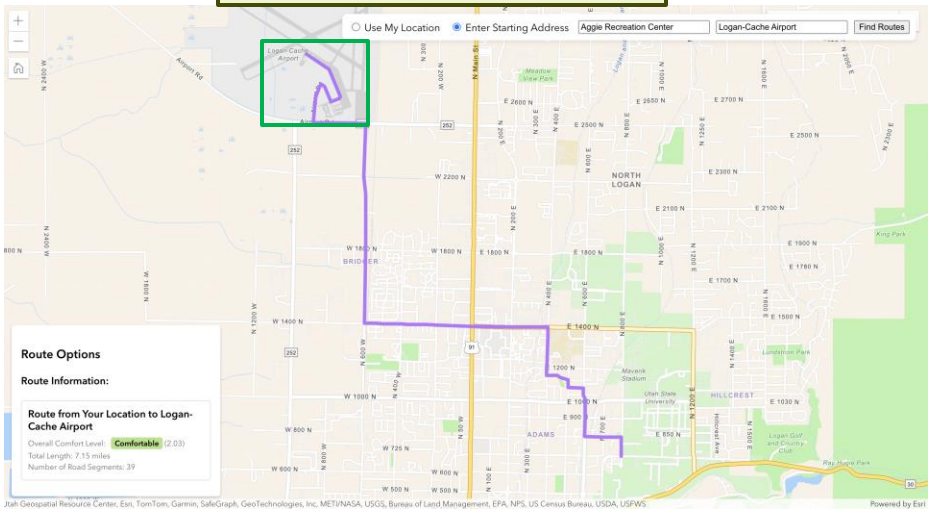
ROUTING DISTANCE COMPARISON



ROUTING DISTANCE COMPARISON

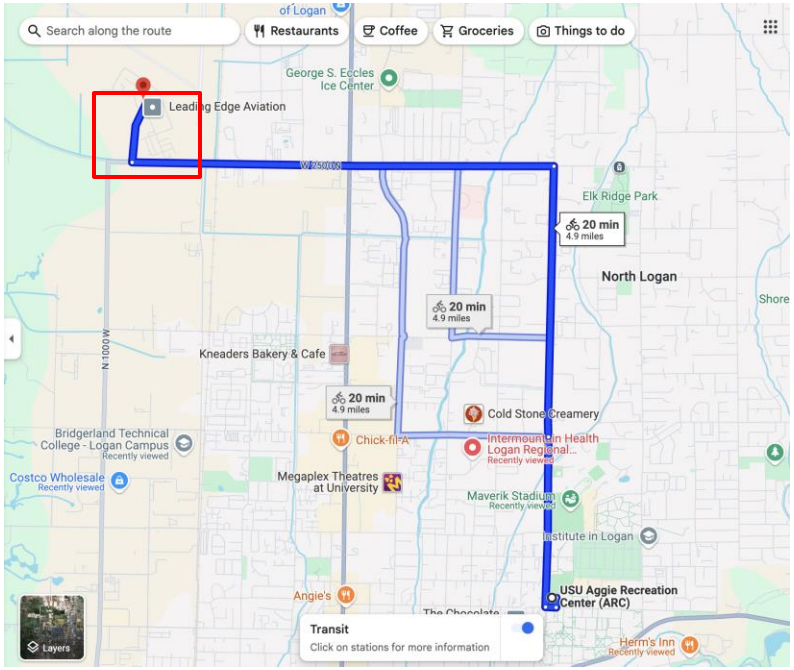
Aggie Recreation Center to Logan-Cache Airport

7.15 miles



Normalized Method

4.7 miles

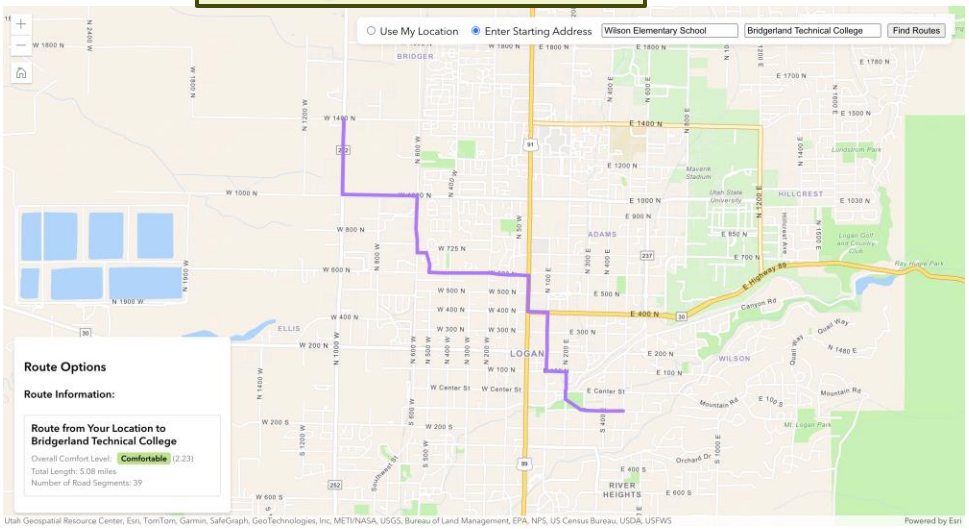


Google Maps

ROUTING DISTANCE COMPARISON

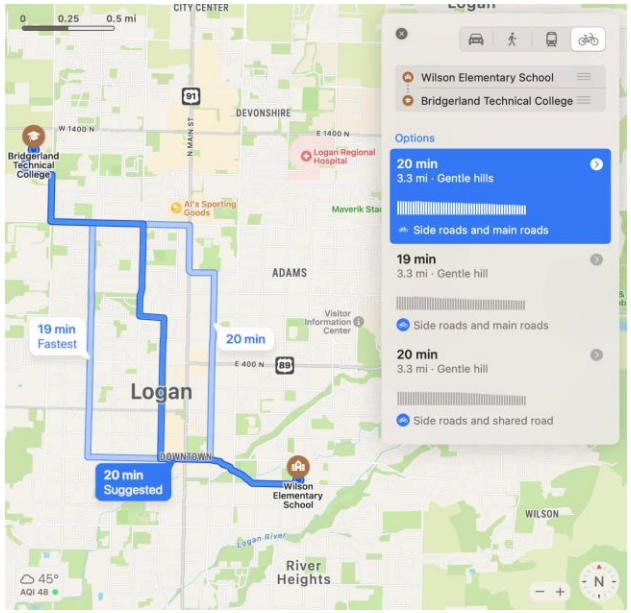
Wilson Elementary School to Bridgerland Technical College

5.08 miles



Normalized Method

3.3 miles

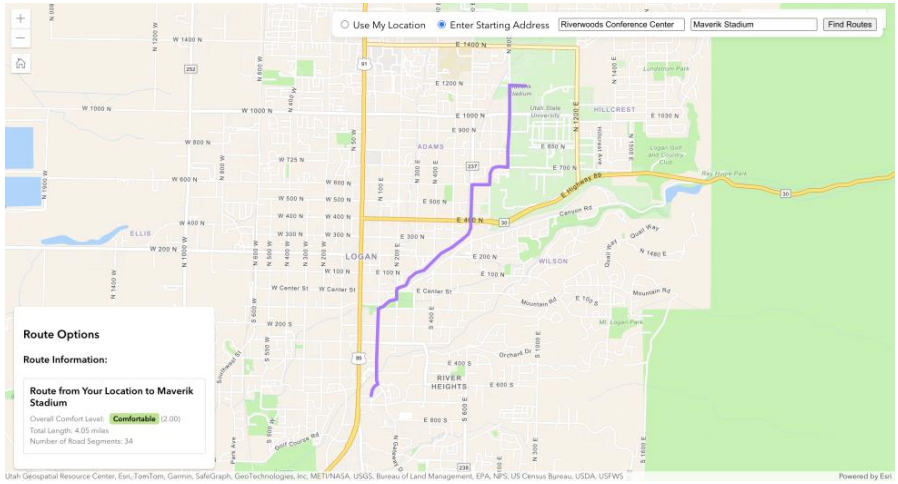


Apple Maps

ROUTING DISTANCE COMPARISON

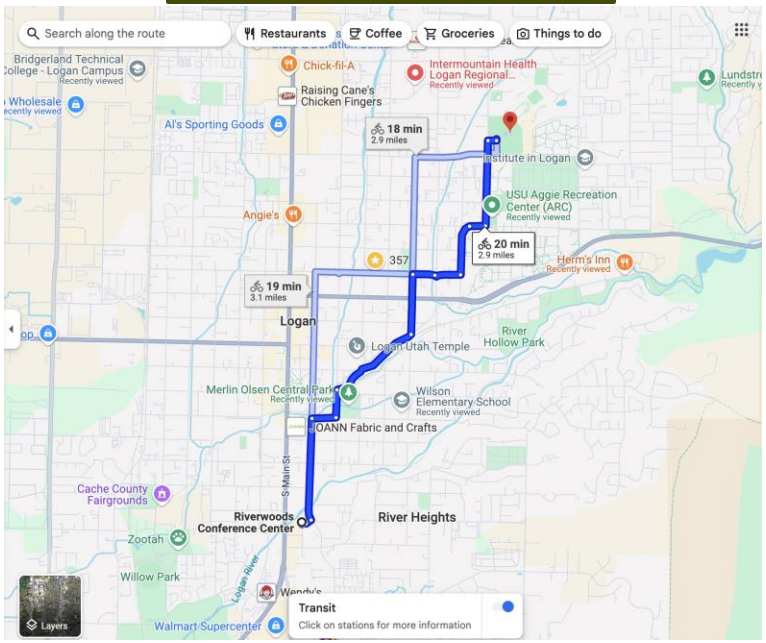
Riverwoods Conference Center to Maverik Stadium

4.05 miles



Additive Method

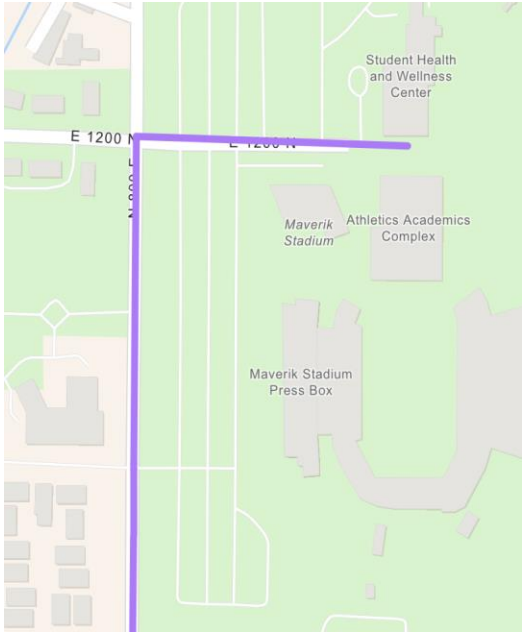
2.9 miles



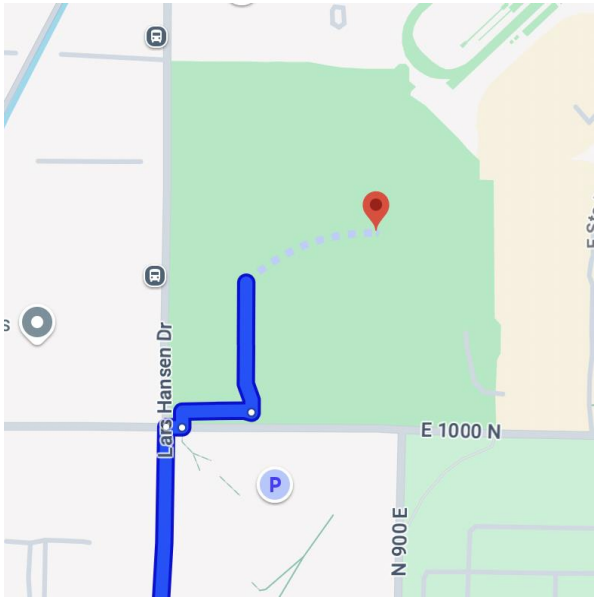
Google Maps

ROUTING DISTANCE COMPARISON

Riverwoods Conference Center to Maverik Stadium – Accounting for Extra Distance



Additive Method



Google Maps

DISCUSSION

Proposed comfortability score based routing – prioritizes trail segments over road and takes into account nonroad paved segments (ie. Parking lots)

Data Gaps / Issues

- Data Updates
- Incorrect Information (WRFC)

Factor Adjustments

- Speed Limit for Trails
- Add in travel lanes?

	Manual Method	Normalized Method
Ease to calibrate		✓
Intuitive		✓
More adjustments	✓	
More consistent	✓	

FUTURE WORK

Activity Data
- Strava

Aggie Blue Bikes
- Survey
- Distribution
- Calibration – QA/QC

- Cycling Comfort vs LTS
- Relative scoring for Cities
- Optimized Routing – Machine Learning

THANK YOU! QUESTIONS?

Web App

Arcpy Toolbox

Alex – a02367780@usu.edu
Eli - a02425459@usu.edu